

TS3000/TS4000

Thickness and Coating Sensor

User Manual version 1.08 202502



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Theory of Operation

The TS3000/TS4000 Thickness and Coating Sensor measures the thickness of glass facing towards the sensor and determines whether the glass is clear or coated (TS3000) or the specific coating type (TS4000). The sensor does not measure the overall thickness of insulated glass units or laminated units. The thickness and coating information are output through an XML interface to an external system (hereafter referred to as the server system).

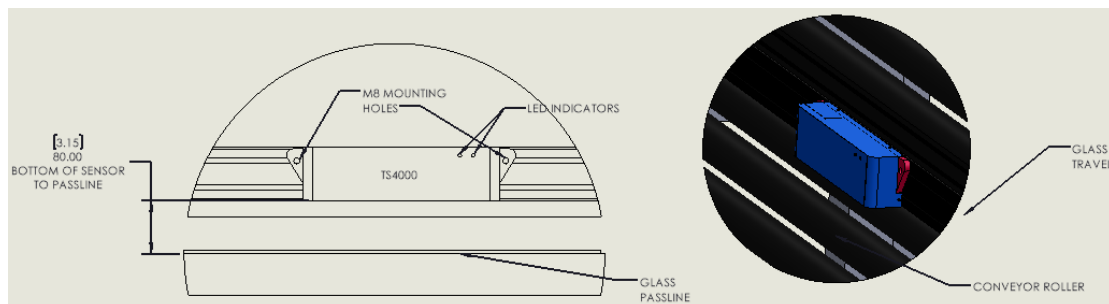
Alignment

The bottom face of the sensor should be mounted parallel to the pass line, with cameras and lights facing down toward the pass line.

Mount the sensor so the bottom face of the sensor is **80mm** from the pass line, top of conveyor roller.

The long dimension of the sensor should be perpendicular to the direction of glass travel. Note that the sensor may be impacted by significant changes in ambient lighting. For best results, maintain consistent surrounding light levels where the sensor is installed.

If needed install a shroud over the top of the sensor extending approximately 100mm longer on both sides. See appendix A.



Network Connection

The sensor communicates via a Cat6 cable connected to whichever device it is configured to communicate with. The sensor is powered via the same Cat6 cable by a 48V POE injector.

Default sensor (client) network configuration:

- IP: 192.168.160.X, where X is the sensor number (1, 2, or 3)
- Port: 42300
- If a custom IP address is used, it can be specified in the "Custom IP" field in the Sensor Selection screen.

Default server network configuration:

- IP: 192.168.160.100

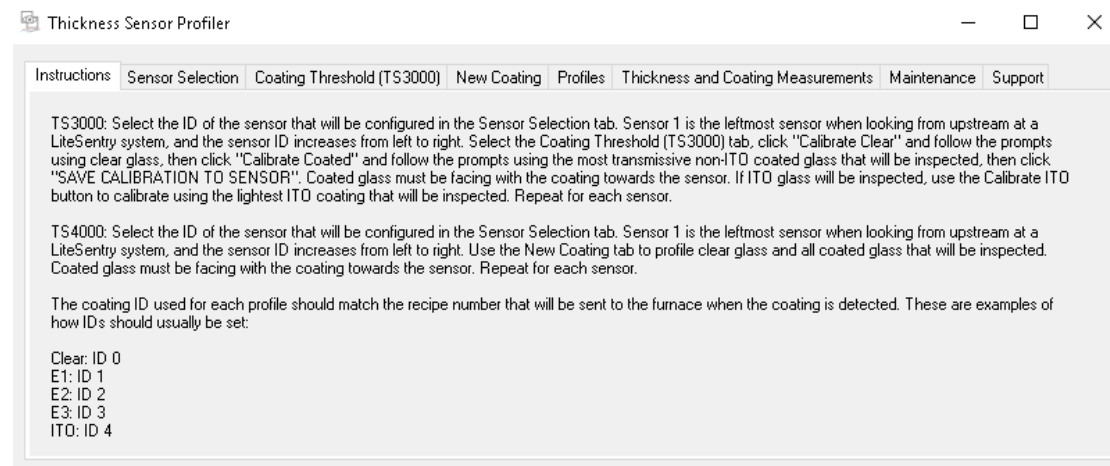
Indicators

- Green LED / Status LED
 - Blinking green indicates the sensor is operational but not communicating with the server system (Osprey System Processor, Owl, etc).
 - Solid green indicates the sensor is on and communicating properly with the server system (Osprey System Processor, Owl, etc).
- Blue LED / Glass Presence LED
 - Blue LED indicates that glass is being detected by the sensor

Thickness Sensor Profiler

The Thickness Sensor Profiler is software to configure the sensor to detect coatings.

On the system that will interact with the sensor, open File Explorer, navigate to C:\LiteSentry\TS Profiler, and run TSX000Profiler.exe. You will see the following screen:



Instructions – gives instructions for configuring sensors.

Sensor Selection – used to set the sensor that is being configured and shows the connection status of each sensor.

Coating Threshold (TS3000) – only present when configuring a TS3000 sensor. This is used to set the transmission threshold between clear and coated glass and to calibrate the sensor for inspecting ITO glass.

New Coating – used for creating coating profiles.

Profiles – used for managing existing coating profiles.

Thickness and Coating Measurements – provides an output of the sensor's current measured thickness, best matching profile, spectral distance to the best matching profile, and whether glass is clear or coated (TS3000 only).

Maintenance – Used by technical support users.

Support – Lists the version of the software and LiteSentry support contact information.

Profiling Glass – *see section below on Profiling Methods.*

For TS3000: Profile any ITO coatings with ID 4.

For TS4000: **Profile Clear first, then all other products with coating top and bottom denoted as mentioned later.**

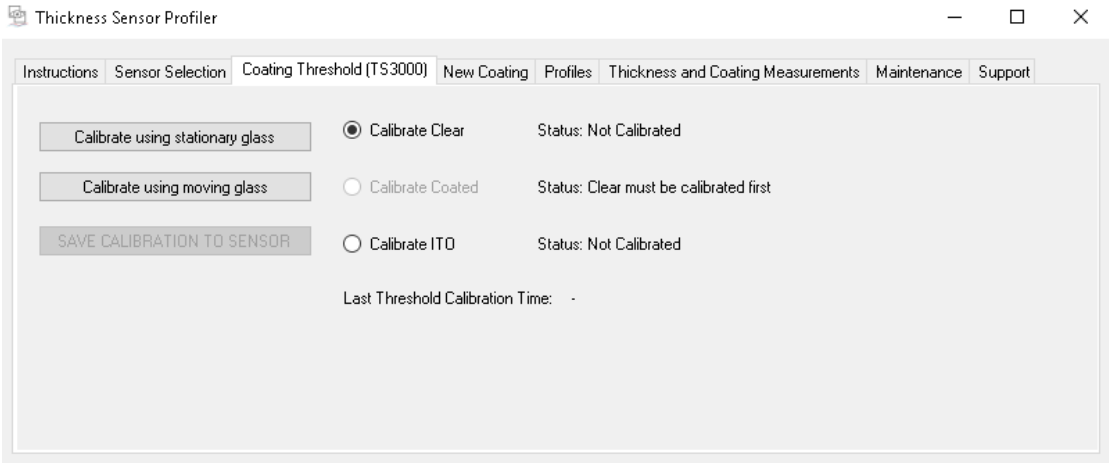
TS3000 & TS4000 first profile to create is Clear Glass

To calibrate the sensor for clear glass, these samples must be available:

Inspection range of thicknesses is greater than 4 mm?	Low iron clear glass will be inspected?	Required Samples
Yes	Yes	• Minimum thickness non-low iron clear glass • Maximum thickness non-low iron clear glass • Minimum thickness low iron clear glass • Maximum thickness low iron clear glass
Yes	No	• Minimum thickness clear glass sample • Maximum thickness clear glass sample
No	Yes	• Low iron clear glass sample • Non-low iron clear glass sample
No	No	• Clear glass sample

Coating Threshold TS3000 (only)

On the system that will interact with the sensor, open File Explorer, navigate to C:\LiteSentry\TS Profiler, and run TSX000Profiler.exe. Navigate to the "Coating Threshold (TS3000)" screen.



Thickness Sensor Profiler

Instructions | Sensor Selection | **Coating Threshold (TS3000)** | New Coating | Profiles | Thickness and Coating Measurements | Maintenance | Support

☒ Calibrate Clear
 Status: Not Calibrated

☐ Calibrate Coated
 Status: Clear must be calibrated first

☐ Calibrate ITO
 Status: Not Calibrated

Last Threshold Calibration Time: -

Tab Definitions:

- **Calibrate Clear** – Select to calibrate the sensor's measurement of clear glass reflection.
- **Calibrate Coated** – Select to calibrate the sensor's measurement of coated glass reflection.
- **Calibrate ITO** – Select to calibrate the sensor's measurement of ITO glass.
- **Calibrate using stationary glass** – the clear or coated glass measurement for calibrating the threshold will be taken using a single spectral measurement, which is taken when the button is clicked.
- **Calibrate using moving glass** – the threshold is calibrated using continuous measurements from when the button is clicked until glass transitions from being present under the sensor to no longer being present under the sensor.
- **Save Threshold Calibration to Sensor** – the threshold is saved to the sensor.

To calibrate the sensor, follow this procedure

1. Select "Calibrate Clear".
2. Select "Calibrate using stationary glass" or "Calibrate using moving glass".
3. The software will ask about the range of thicknesses of clear glass that the sensor will inspect in operation and whether low iron glass will be inspected, then a list of the required samples will be shown. If the required samples shown are available, click "OK".
4. Follow the remaining prompts.
5. Select "Calibrate Coated".
6. Repeat Step 2 and Step 4 using a sample of the most transmissive silver based, non-ITO glass that should be detected as coated glass.
7. Select "Save Coating Threshold Calibration to Sensor". When a message appears, click "OK".

The following steps are only necessary if inspecting ITO glass.

8. Select "Calibrate ITO".
9. Repeat Step 2 and Step 4 using a sample of ITO glass.

In the Sensor Selection screen, set the Selected Sensor ID option to the sensor number of the sensor. This can be found on the label below the Ethernet port on the sensor.



Coating Profiles TS4000

Go back to the **New Coating** screen.

- Enter the name of the profile in the Name box
- Enter an ID number of the profile in the ID box.
 - If the coating is on the bottom side of the glass, use any negative number for the ID.
 - If the coating is on the top side of the glass, use the furnace recipe number for this coating as the ID.

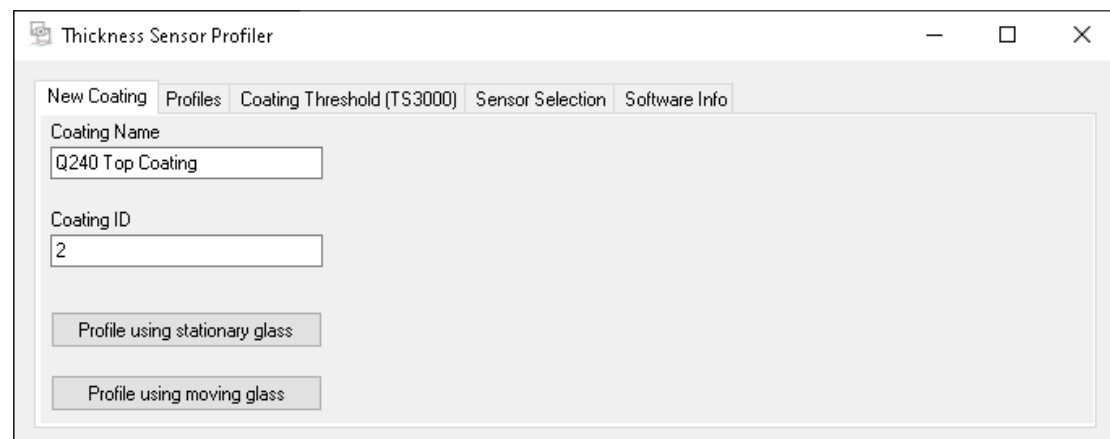
Example:

Coating Name: Q240 Top Coating

Coating ID: 2

Coating Name: Q240 Bottom Coating

Coating ID: -2



Profiling Methods

Products can be profiled using stationery or moving samples.

Profiling using stationary glass

Click "Profile using stationary glass" and follow the on-screen instructions.

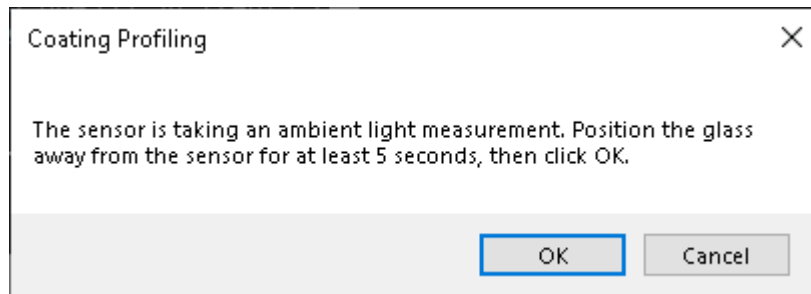
NOTE: The system's lightbox lights must be turned on for the rest of this process in order to best simulate the actual inspection environment. A message will appear to warn you of this. This can be achieved by free running the system or bypassing the lightbox relay inside of the electrical console.

✕

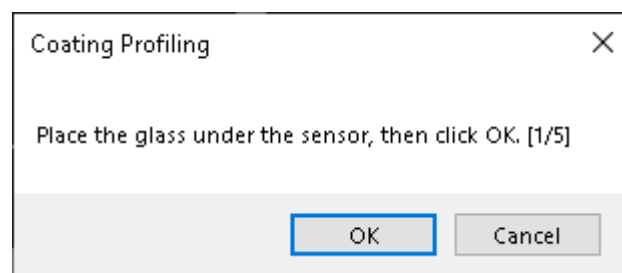
Turn the lightbox lights on, then click OK. On LiteSentry systems, the lightbox lights can be turned on by overriding the lightbox relay in the electrical cabinet.

OK

Ensure the test glass piece is not positioned under the sensor for at least 3 seconds to ensure proper capture of ambient light conditions, then click OK.



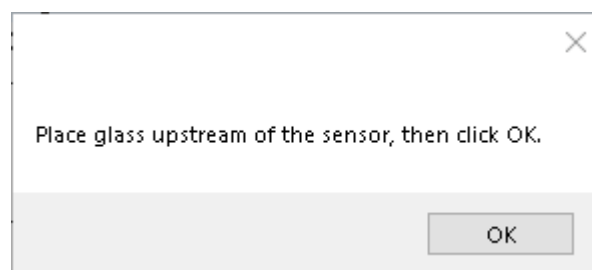
Place the test glass piece under the sensor and Click OK to capture the first of five total samples.

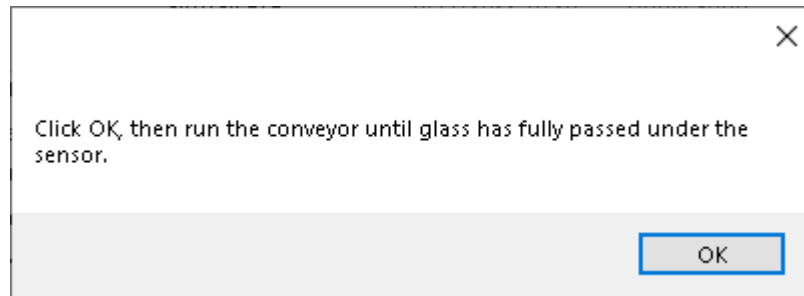


Move the glass to a different position under the sensor, then click OK. Repeat this for five different locations. After five measurements of glass have been taken, the distance metric between the current measurement and the new profile will be shown. If the distance is greater than 50, the profile should be deleted by following the instructions for deleting a profile in the "Managing Profiles" section, and the glass should be reprofiled.

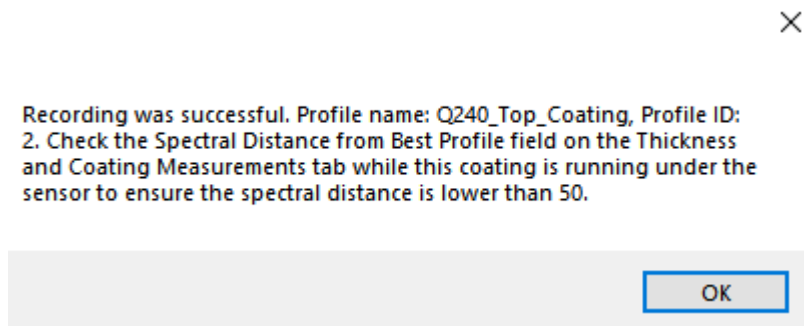
Profiling using moving glass

Click "Profile using moving glass". Follow the prompts that appear.





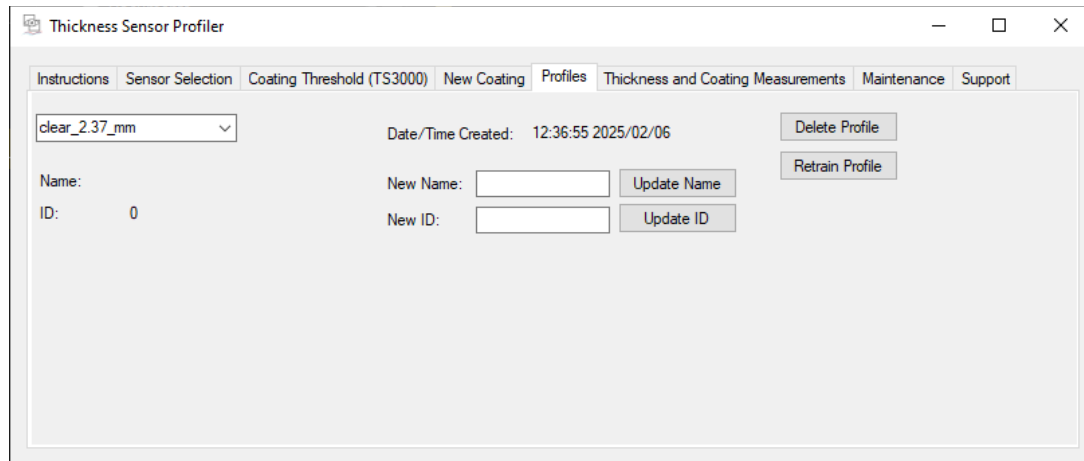
After glass has fully passed under the sensor, a message will indicate that the profile was created.



A new profile will be saved to the sensor based on its spectral measurements taken while glass was running under the sensor.

Managing Profiles

The "Profiles" screen lists information about profiles previously created. The information for a specific profile can be viewed by selecting the name of the profile from the drop-down menu.



Retrain Profile – the currently selected profile will be retrained using either stationary or moving glass, updating its data with new measurements.

Delete Profile – the currently selected profile will be deleted from the sensor.

Update Name – the name of the currently selected profile will be updated to the new name entered.

Update ID – the ID of the currently selected profile will be updated to the new ID entered.

To retrain an existing profile:

1. Select the desired profile from the drop-down menu.
2. Click "Retrain Profile".
3. Choose either yes for "Retrain using stationary glass" or no for "Retrain using moving glass".
4. Follow the on-screen prompts to capture new measurements for the selected profile.
5. The profile will be updated with the new data, and a confirmation message will be displayed.

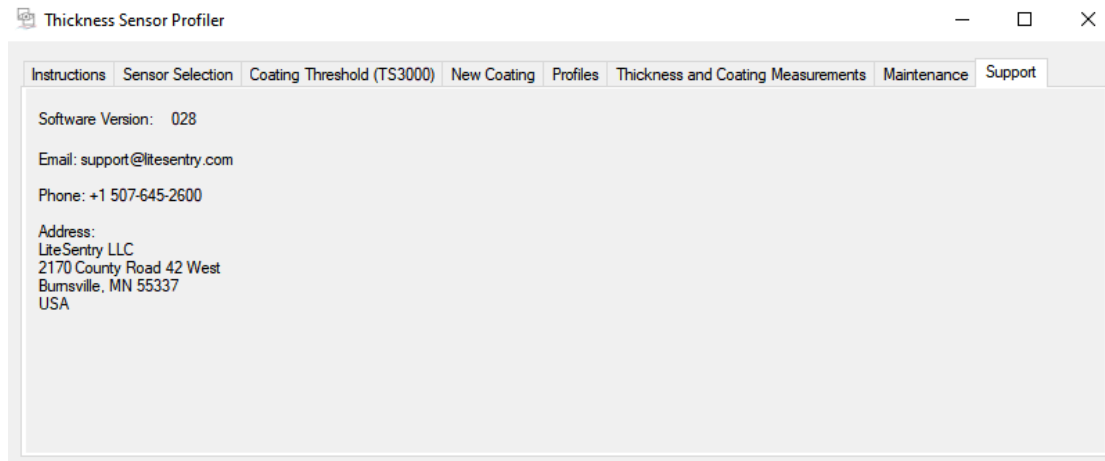
Option for misidentified coating:

If a coating is not properly identified there is the option of ADDING a profile using the same coating ID.

Go to New Coating tab, create a profile, use the same name with the addition of a suffix letter or number, use the same ID as the original coating.

Technical Support Information

The Support screen shows the version number of the Profiler software and lists contact information for LiteSentry support.



Appendix A

Shade Plate dimensions for standard TSx sensor mounted to 80x80 extrusion.

